PLAY, LEARN & PROTECT

Technology for Childhood

18 Proposed Bachelor Graduation Project Topics

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PROTECT — Digital Safety

1. AI-Based Detection of Misinformation and Fake Content Targeting Children and Families

Description

Children and parents increasingly encounter misleading health claims, safety hoaxes, and fabricated "viral challenge" content on platforms aimed at young audiences. This project builds a text-classification model to flag misinformation in child- and family-facing content, following the CRISP-DM approach used across this research group: source a suitable public misinformation/fake-news dataset, clean and label it (adapting labels to the child-safety context where needed), and apply standard classification algorithms (Logistic Regression, Random Forest, SVM, and a pretrained transformer classifier such as BERT) to compare performance. The emphasis is on careful data preparation, model comparison, and a clear, well-justified write-up rather than novel algorithm design.

Prerequisites

Basic Python (the level covered in the department's introductory programming and data science courses) and familiarity with scikit-learn; no deep learning background required — pretrained models are used off the shelf via simple library calls, not built or trained from scratch.

Proposed Workplan (14 Weeks)

Phase / Weeks	Focus	Key Activities
Phase 1 — Weeks 1–2	Scoping & Literature Review	Define research question and objectives; review academic literature and existing systems/products; finalize scope with supervisor sign-off.
Phase 2 — Weeks 3–4	Dataset Sourcing, Cleaning & Labeling	Identify and source datasets (or design data-collection protocol); complete data cleaning, exploratory analysis, and ethical/privacy review where child data is involved.
Phase 3 — Weeks 5–8	Model Building & Comparison (Classical ML + Pretrained Classifier)	Design and build the core model/system/prototype; implement baseline, then iterate; document design decisions and trade-offs.
Phase 4 — Weeks 9–11	Evaluation & Refinement	Test against defined metrics/benchmarks or original reference paper; run comparative analysis; refine based on results; conduct usability or expert validation where applicable.
Phase 5 — Weeks 12–13	Documentation	Write up methodology, results, and discussion; prepare figures/tables; complete limitations and future-work sections.
Phase 6 — Week 14	Defense Preparation	Prepare presentation and demo; rehearse defense; incorporate supervisor feedback from final review.

2. Evaluating AI Tools for Detecting Manipulated Images and Videos on Child-Facing Platforms

Description

As AI image and video generation tools become widely accessible, child-facing platforms face growing exposure to manipulated media. Rather than building a detection model from scratch — a task requiring specialized computer-vision expertise — this project takes an applied, evaluative approach: the student will identify and test several existing, freely available deepfake-detection tools and APIs (e.g. open-source detectors, cloud vision APIs) against a curated test set of real and manipulated images/videos, measuring accuracy, ease of integration, and cost. The output is a practical vendor/tool comparison a real product team could use to choose a detection solution, rather than a custom-built detector.

Prerequisites

No computer-vision or deep-learning background required; basic Python knowledge is sufficient.

Proposed Workplan (14 Weeks)

Phase / Weeks	Focus	Key Activities
Phase 1 — Weeks 1–2	Scoping & Literature Review	Define research question and objectives; review academic literature and existing systems/products; finalize scope with supervisor sign-off.
Phase 2 — Weeks 3–4	Test-Set Assembly & Candidate Tool Shortlisting	Identify and source datasets (or design data-collection protocol); complete data cleaning, exploratory analysis, and ethical/privacy review where child data is involved.
Phase 3 — Weeks 5–8	Tool Testing & Benchmark Scoring	Design and build the core model/system/prototype; implement baseline, then iterate; document design decisions and trade-offs.
Phase 4 — Weeks 9–11	Evaluation & Refinement	Test against defined metrics/benchmarks or original reference paper; run comparative analysis; refine based on results; conduct usability or expert validation where applicable.
Phase 5 — Weeks 12–13	Documentation	Write up methodology, results, and discussion; prepare figures/tables; complete limitations and future-work sections.
Phase 6 — Week 14	Defense Preparation	Prepare presentation and demo; rehearse defense; incorporate supervisor feedback from final review.

3. AI-Driven Age Verification and Estimation for Digital Platforms

Description

Age verification sits at the center of current global regulatory debate (COPPA, the EU's DSA, UK Online Safety Act) and is directly relevant to any product targeting children. This project compares three approaches — off-the-shelf facial age-estimation APIs, simple behavioral-signal scoring (e.g. rule-based scoring from account activity patterns), and self-declaration with verification friction — evaluating each on accuracy, fairness across demographic groups, and privacy cost. The student will use existing pretrained age-estimation APIs rather than building a model, and focus the bulk of the effort on the comparative research and decision framework.

Prerequisites

Basic machine learning background required. Interest in the policy/regulatory dimension is valuable since the deliverable includes a compliance framework.

Proposed Workplan (14 Weeks)

Phase / Weeks	Focus	Key Activities
Phase 1 — Weeks 1–2	Scoping & Literature Review	Define research question and objectives; review academic literature and existing systems/products; finalize scope with supervisor sign-off.
Phase 2 — Weeks 3–4	Method Selection & Proxy-Signal/API Access Setup	Identify and source datasets (or design data-collection protocol); complete data cleaning, exploratory analysis, and ethical/privacy review where child data is involved.
Phase 3 — Weeks 5–8	Comparative Testing Using Existing APIs	Design and build the core model/system/prototype; implement baseline, then iterate; document design decisions and trade-offs.
Phase 4 — Weeks 9–11	Evaluation & Refinement	Test against defined metrics/benchmarks or original reference paper; run comparative analysis; refine based on results; conduct usability or expert validation where applicable.
Phase 5 — Weeks 12–13	Documentation	Write up methodology, results, and discussion; prepare figures/tables; complete limitations and future-work sections.
Phase 6 — Week 14	Defense Preparation	Prepare presentation and demo; rehearse defense; incorporate supervisor feedback from final review.

4. Predictive Risk Scoring for Livestream and Short-Video Chat Comments

Description

Livestream and short-video platforms present a moderation challenge because chat and comments arrive faster than moderators can read them. This project builds a text-based risk-scoring model — using standard classification techniques (Logistic Regression, Random Forest, or a pretrained NLP classifier) applied to publicly available livestream/comment chat-log datasets — that flags messages showing harassment, coordinated abuse, or spam/scam patterns. The model is kept interpretable (using simple feature importance, not black-box deep learning) so that a non-technical moderator can understand why a message was flagged.

Prerequisites

Basic Python and familiarity with scikit-learn; no audio processing or multimodal ML required.

Proposed Workplan (14 Weeks)

Phase / Weeks	Focus	Key Activities
Phase 1 — Weeks 1–2	Scoping & Literature Review	Define research question and objectives; review academic literature and existing systems/products; finalize scope with supervisor sign-off.
Phase 2 — Weeks 3–4	Chat/Comment Dataset Assembly & Labeling	Identify and source datasets (or design data-collection protocol); complete data cleaning, exploratory analysis, and ethical/privacy review where child data is involved.
Phase 3 — Weeks 5–8	Risk Classification Model Development	Design and build the core model/system/prototype; implement baseline, then iterate; document design decisions and trade-offs.
Phase 4 — Weeks 9–11	Evaluation & Refinement	Test against defined metrics/benchmarks or original reference paper; run comparative analysis; refine based on results; conduct usability or expert validation where applicable.
Phase 5 — Weeks 12–13	Documentation	Write up methodology, results, and discussion; prepare figures/tables; complete limitations and future-work sections.
Phase 6 — Week 14	Defense Preparation	Prepare presentation and demo; rehearse defense; incorporate supervisor feedback from final review.

5. Auditing Exploitative Design Patterns ("Dark Patterns") in Apps Targeting Children

Description

Beyond content risk, many child-facing apps embed manipulative design — forced-continuity subscriptions, disguised ads, and reward loops engineered for compulsive engagement. This project is structured as a systematic audit rather than a technical build: the student develops a scoring checklist from the published dark-pattern taxonomy literature (e.g. Mathur et al.), manually reviews a sample of 20–30 popular children's apps against the checklist, and uses spreadsheet-based scoring and visualization (Excel or Power BI) to identify the most common manipulative patterns and which app categories are most affected.

Prerequisites

No coding required; strong attention to detail and structured research skills; experience with Excel/Power BI for organizing and visualizing audit findings.

Proposed Workplan (14 Weeks)

Phase / Weeks	Focus	Key Activities
Phase 1 — Weeks 1–2	Scoping & Literature Review	Review the dark-pattern taxonomy literature; define the audit checklist criteria; finalize scope with supervisor sign-off.
Phase 2 — Weeks 3–5	App Sample Selection & Checklist Refinement	Select 20–30 popular children's apps across categories (games, education, social); pilot-test the checklist on a small subset and refine scoring definitions.
Phase 3 — Weeks 6–9	Systematic Audit & Data Collection	Conduct the full audit across the app sample; document evidence (screenshots, notes) for each flagged pattern; compile findings into a structured spreadsheet.
Phase 4 — Weeks 10–11	Analysis & Visualization	Analyze patterns across app categories using Excel/Power BI; identify the most prevalent and most harmful design patterns; validate findings against the taxonomy literature.
Phase 5 — Weeks 12–13	Documentation	Write up methodology, audit findings, and a proposed "child-safe by design" scoring rubric; prepare supporting charts and tables.
Phase 6 — Week 14	Defense Preparation	Prepare presentation; rehearse defense; incorporate supervisor feedback.

6. AI-Based Anomaly Detection for Suspicious Account Activity on Child-Facing Platforms

Description

Unusual account behavior — a sudden change in messaging patterns, contact requests from many new accounts in a short window, or activity at unusual hours — can be an early indicator of risk. This project applies standard anomaly-detection techniques (Isolation Forest, One-Class SVM, or simple statistical outlier detection) to structured account-activity data (login times, message frequency, contact-request patterns) to flag accounts worth a closer look, following the same classical-ML, CRISP-DM-based workflow.

Prerequisites

Basic Python and scikit-learn familiarity, at the same level as standard classification/clustering coursework; ML background needed.

Proposed Workplan (14 Weeks)

Phase / Weeks	Focus	Key Activities
Phase 1 — Weeks 1–2	Scoping & Literature Review	Define research question and objectives; review academic literature and existing systems/products; finalize scope with supervisor sign-off.
Phase 2 — Weeks 3–4	Structured Activity Dataset Sourcing & Feature Engineering	Identify and source datasets (or design data-collection protocol); complete data cleaning, exploratory analysis, and ethical/privacy review where child data is involved.
Phase 3 — Weeks 5–8	Anomaly Detection Model Development & Threshold Tuning	Design and build the core model/system/prototype; implement baseline, then iterate; document design decisions and trade-offs.
Phase 4 — Weeks 9–11	Evaluation & Refinement	Test against defined metrics/benchmarks or original reference paper; run comparative analysis; refine based on results; conduct usability or expert validation where applicable.
Phase 5 — Weeks 12–13	Documentation	Write up methodology, results, and discussion; prepare figures/tables; complete limitations and future-work sections.
Phase 6 — Week 14	Defense Preparation	Prepare presentation and demo; rehearse defense; incorporate supervisor feedback from final review.

LEARN — Personalized & Safe Education

7. Rule-Based and Predictive Adaptive Difficulty Engines for Educational Games

Description

Educational games succeed or fail on pacing: too easy and children disengage, too hard and they quit in frustration. This project designs a difficulty-adjustment system driven by a simple predictive model (e.g. Logistic Regression or Decision Tree trained on player performance data — accuracy, response time, retry rate) that recommends the next difficulty level, compared against a static-difficulty baseline and a simple rule-based baseline (e.g. "increase difficulty after 3 correct answers in a row"). This keeps the modeling approach within standard classification/prediction methods already used successfully across this research group, rather than requiring reinforcement learning.

Prerequisites

Basic Python and scikit-learn; a simple existing game template or a no-code/low-code game-builder tool (e.g. Construct, Scratch-level web tools) can be used for the testbed game, since the focus of the project is the difficulty-prediction model, not game engineering.

Proposed Workplan (14 Weeks)

Phase / Weeks	Focus	Key Activities
Phase 1 — Weeks 1–2	Scoping & Literature Review	Define research question and objectives; review academic literature and existing systems/products; finalize scope with supervisor sign-off.
Phase 2 — Weeks 3–4	Testbed Game Setup & Performance Data Design	Identify and source datasets (or design data-collection protocol); complete data cleaning, exploratory analysis, and ethical/privacy review where child data is involved.
Phase 3 — Weeks 5–8	Difficulty Prediction Model Development	Design and build the core model/system/prototype; implement baseline, then iterate; document design decisions and trade-offs.
Phase 4 — Weeks 9–11	Evaluation & Refinement	Test against defined metrics/benchmarks or original reference paper; run comparative analysis; refine based on results; conduct usability or expert validation where applicable.
Phase 5 — Weeks 12–13	Documentation	Write up methodology, results, and discussion; prepare figures/tables; complete limitations and future-work sections.
Phase 6 — Week 14	Defense Preparation	Prepare presentation and demo; rehearse defense; incorporate supervisor feedback from final review.

8. Multilingual (Arabic-English) NLP for Children's Reading Level Assessment

Description

Reading-level assessment tools are overwhelmingly built and validated on English text, leaving a real gap for Arabic-language EdTech — directly relevant to localizing Play, Learn & Protect for the Egyptian/MENA market. This project builds a readability classifier for Arabic and English children's text by combining well-established readability formulas (e.g. Flesch-Kincaid for English, and simple Arabic-adapted metrics such as sentence length and vocabulary-frequency bands) with a pretrained Arabic transformer classifier (AraBERT).

Prerequisites

Basic Python and comfort with Hugging Face's simple fine-tuning workflow (well-documented and tutorial-supported, not building a model architecture from scratch).

Proposed Workplan (14 Weeks)

Phase / Weeks	Focus	Key Activities
Phase 1 — Weeks 1–2	Scoping & Literature Review	Define research question and objectives; review academic literature and existing systems/products; finalize scope with supervisor sign-off.
Phase 2 — Weeks 3–4	Corpus Assembly (Arabic & English Children's Texts) & Readability Labeling	Identify and source datasets (or design data-collection protocol); complete data cleaning, exploratory analysis, and ethical/privacy review where child data is involved.
Phase 3 — Weeks 5–8	Classifier Development (AraBERT Fine-Tuning + Readability Formulas)	Design and build the core model/system/prototype; implement baseline, then iterate; document design decisions and trade-offs.
Phase 4 — Weeks 9–11	Evaluation & Refinement	Test against defined metrics/benchmarks or original reference paper; run comparative analysis; refine based on results; conduct usability or expert validation where applicable.
Phase 5 — Weeks 12–13	Documentation	Write up methodology, results, and discussion; prepare figures/tables; complete limitations and future-work sections.
Phase 6 — Week 14	Defense Preparation	Prepare presentation and demo; rehearse defense; incorporate supervisor feedback from final review.

9. Designing Safety Guardrails for AI Tutoring Chatbots for Children

Description

AI chatbots are increasingly used for tutoring, but general-purpose safety settings are not designed with children in mind. This project is primarily a design and evaluation study rather than a model-building exercise: using an existing chatbot platform or LLM API (e.g. via simple, well-documented API calls or a no-code chatbot builder), the student designs a set of content rules and conversation boundaries appropriate for children, then systematically tests the chatbot with a structured set of edge-case questions to see where it needs improvement, iterating on the rules based on results.

Prerequisites

No advanced coding required; comfort using APIs at a basic level, or a no-code chatbot-builder tool; interest in child-safety/communication design is valuable since guardrail wording is a core part of the deliverable.

Proposed Workplan (14 Weeks)

Phase / Weeks	Focus	Key Activities
Phase 1 — Weeks 1–2	Scoping & Literature Review	Define research question and objectives; review academic literature and existing systems/products; finalize scope with supervisor sign-off.
Phase 2 — Weeks 3–4	Guardrail Requirements Definition & Baseline Chatbot Setup	Identify and source datasets (or design data-collection protocol); complete data cleaning, exploratory analysis, and ethical/privacy review where child data is involved.
Phase 3 — Weeks 5–8	Guardrail Design & Structured Testing	Design and build the core model/system/prototype; implement baseline, then iterate; document design decisions and trade-offs.
Phase 4 — Weeks 9–11	Evaluation & Refinement	Test against defined metrics/benchmarks or original reference paper; run comparative analysis; refine based on results; conduct usability or expert validation where applicable.
Phase 5 — Weeks 12–13	Documentation	Write up methodology, results, and discussion; prepare figures/tables; complete limitations and future-work sections.
Phase 6 — Week 14	Defense Preparation	Prepare presentation and demo; rehearse defense; incorporate supervisor feedback from final review.

10. Gamified Cyber Education: Measuring Behavior Change

Description

Digital-literacy curricula often teach safety rules that don't translate into changed behavior. This project designs a short gamified cyber-awareness module (password hygiene, phishing recognition, oversharing awareness) — which can be built using a no-code quiz/game platform rather than custom software — and measures whether in-game learning transfers to real behavioral change, using a simple pre-test/post-test experimental design and standard statistical comparison (t-tests, in Excel or SPSS).

Prerequisites

Basic coding & ML knowledge required; a no-code quiz/game-builder tool (e.g. Kahoot-style platforms, Google Forms-based interactive modules) is used for a start; basic statistics helps with the pre/post comparison analysis.

Proposed Workplan (14 Weeks)

Phase / Weeks	Focus	Key Activities
Phase 1 — Weeks 1–2	Scoping & Literature Review	Define research question and objectives; review academic literature and existing systems/products; finalize scope with supervisor sign-off.
Phase 2 — Weeks 3–4	Curriculum & Game Module Design (No-Code Tools)	Identify and source datasets (or design data-collection protocol); complete data cleaning, exploratory analysis, and ethical/privacy review where child data is involved.
Phase 3 — Weeks 5–8	Behavioral Transfer Study Design & Data Collection	Design and build the core model/system/prototype; implement baseline, then iterate; document design decisions and trade-offs.
Phase 4 — Weeks 9–11	Evaluation & Refinement	Test against defined metrics/benchmarks or original reference paper; run comparative analysis; refine based on results; conduct usability or expert validation where applicable.
Phase 5 — Weeks 12–13	Documentation	Write up methodology, results, and discussion; prepare figures/tables; complete limitations and future-work sections.
Phase 6 — Week 14	Defense Preparation	Prepare presentation and demo; rehearse defense; incorporate supervisor feedback from final review.

11. AI-Based Skill Assessment Through Gameplay Data

Description

Traditional tests are a blunt instrument for measuring skills like problem-solving in children. This project explores whether simple in-game action logs — number of attempts, time-to-solution, hint usage — can serve as a signal of skill level, following the same feature-engineering-plus-classical-ML pattern used in the existing behavior-mapping theses in this research group (e.g. Random Forest or clustering on engineered features), rather than requiring custom game instrumentation or advanced telemetry pipelines. A publicly available educational-game dataset (e.g. from Kaggle or an open EdTech research repository) can be used in place of building a new game from scratch.

Prerequisites

Basic Python and scikit-learn, at the level of standard classification/clustering coursework; no custom software development required if using an existing public dataset.

Proposed Workplan (14 Weeks)

Phase / Weeks	Focus	Key Activities
Phase 1 — Weeks 1–2	Scoping & Literature Review	Define research question and objectives; review academic literature and existing systems/products; finalize scope with supervisor sign-off.
Phase 2 — Weeks 3–4	Dataset Sourcing (Existing Gameplay Data) & Feature Engineering	Identify and source datasets (or design data-collection protocol); complete data cleaning, exploratory analysis, and ethical/privacy review where child data is involved.
Phase 3 — Weeks 5–8	Skill Prediction Model Development & Validation	Design and build the core model/system/prototype; implement baseline, then iterate; document design decisions and trade-offs.
Phase 4 — Weeks 9–11	Evaluation & Refinement	Test against defined metrics/benchmarks or original reference paper; run comparative analysis; refine based on results; conduct usability or expert validation where applicable.
Phase 5 — Weeks 12–13	Documentation	Write up methodology, results, and discussion; prepare figures/tables; complete limitations and future-work sections.
Phase 6 — Week 14	Defense Preparation	Prepare presentation and demo; rehearse defense; incorporate supervisor feedback from final review.

12. Explainable, No-Code Parent-Facing Learning Progress Dashboards

Description

Parents are frequently handed opaque "progress scores" from EdTech platforms with no explanation of what drives them. This project takes an existing simple classification/regression model (built with standard scikit-learn tools, and applies a beginner-friendly interpretability library (SHAP, which has a simple, well-documented plotting interface) to translate model outputs into plain-language insights, then builds the parent-facing dashboard using a no-code/low-code BI tool (Power BI) rather than custom front-end development.

Prerequisites

Basic Python/scikit-learn for the underlying model; Power BI proficiency for the dashboard (in line with the no-code tools already emphasized in this group's business-education work).

Proposed Workplan (14 Weeks)

Phase / Weeks	Focus	Key Activities
Phase 1 — Weeks 1–2	Scoping & Literature Review	Define research question and objectives; review academic literature and existing systems/products; finalize scope with supervisor sign-off.
Phase 2 — Weeks 3–4	Learning Model Selection/Training & SHAP Setup	Identify and source datasets (or design data-collection protocol); complete data cleaning, exploratory analysis, and ethical/privacy review where child data is involved.
Phase 3 — Weeks 5–8	Dashboard Design in Power BI/Tableau	Design and build the core model/system/prototype; implement baseline, then iterate; document design decisions and trade-offs.
Phase 4 — Weeks 9–11	Evaluation & Refinement	Test against defined metrics/benchmarks or original reference paper; run comparative analysis; refine based on results; conduct usability or expert validation where applicable.
Phase 5 — Weeks 12–13	Documentation	Write up methodology, results, and discussion; prepare figures/tables; complete limitations and future-work sections.
Phase 6 — Week 14	Defense Preparation	Prepare presentation and demo; rehearse defense; incorporate supervisor feedback from final review.

PLAY — Safe, Engaging Digital Play

13. AI-Generated Personalized Storytelling for Child Engagement and Values Education

Description

This project uses an existing generative AI tool (via simple API calls or a no-code AI writing platform) to produce personalized, culturally localized stories that reinforce safety and values lessons — for example, a story featuring the child's own name and interests that embeds a lesson about not sharing personal information online. The core contribution is designing a clear content-safety checklist and review workflow (prompt constraints, a human review step, and a keyword/topic filter built with simple rule-based logic) rather than any custom model development.

Prerequisites

Basic coding skills; comfort using a generative AI API or platform at a basic level; interest in content design and child development.

Proposed Workplan (14 Weeks)

Phase / Weeks	Focus	Key Activities
Phase 1 — Weeks 1–2	Scoping & Literature Review	Define research question and objectives; review academic literature and existing systems/products; finalize scope with supervisor sign-off.
Phase 2 — Weeks 3–4	Story Framework & Safety Checklist Design	Identify and source datasets (or design data-collection protocol); complete data cleaning, exploratory analysis, and ethical/privacy review where child data is involved.
Phase 3 — Weeks 5–8	Generation Workflow Setup & Filter Testing	Design and build the core model/system/prototype; implement baseline, then iterate; document design decisions and trade-offs.
Phase 4 — Weeks 9–11	Evaluation & Refinement	Test against defined metrics/benchmarks or original reference paper; run comparative analysis; refine based on results; conduct usability or expert validation where applicable.
Phase 5 — Weeks 12–13	Documentation	Write up methodology, results, and discussion; prepare figures/tables; complete limitations and future-work sections.
Phase 6 — Week 14	Defense Preparation	Prepare presentation and demo; rehearse defense; incorporate supervisor feedback from final review.

14. Recommendation Engines for Age-Appropriate Content Curation

Description

Standard recommendation engines optimize for engagement, which in adult contexts drives the well-documented "rabbit-hole" effect — an outcome actively undesirable in a children's product. This project builds a simple collaborative-filtering or content-based recommender (using standard, well-documented Python libraries such as Surprise or scikit-learn, that adds basic diversity and session-length constraints, and compares it against a standard engagement-maximizing baseline on relevance and "healthy usage" metrics.

Prerequisites

Basic Python; recommender-systems concepts covered in standard coursework (collaborative filtering, content-based filtering).

Proposed Workplan (14 Weeks)

Phase / Weeks	Focus	Key Activities
Phase 1 — Weeks 1–2	Scoping & Literature Review	Define research question and objectives; review academic literature and existing systems/products; finalize scope with supervisor sign-off.
Phase 2 — Weeks 3–4	Dataset Preparation & Baseline Recommender Development	Identify and source datasets (or design data-collection protocol); complete data cleaning, exploratory analysis, and ethical/privacy review where child data is involved.
Phase 3 — Weeks 5–8	Constrained Recommender Development (Diversity & Session-Cap Rules)	Design and build the core model/system/prototype; implement baseline, then iterate; document design decisions and trade-offs.
Phase 4 — Weeks 9–11	Evaluation & Refinement	Test against defined metrics/benchmarks or original reference paper; run comparative analysis; refine based on results; conduct usability or expert validation where applicable.
Phase 5 — Weeks 12–13	Documentation	Write up methodology, results, and discussion; prepare figures/tables; complete limitations and future-work sections.
Phase 6 — Week 14	Defense Preparation	Prepare presentation and demo; rehearse defense; incorporate supervisor feedback from final review.

15. Predicting Problematic Screen-Time Patterns from Usage Data

Description

Most screen-time tools simply track and cap usage after the fact. This project instead predicts problematic usage trends using standard classification techniques (Logistic Regression, Random Forest) applied to simple usage-summary features (average daily session count, time-of-day patterns, week-over-week trend) rather than complex time-series modeling, keeping the technical approach consistent with the classical-ML work already completed across this research group. The student will work with simulated or publicly available app-usage datasets.

Prerequisites

Basic Python and scikit-learn; no time-series-specific modeling background required, since usage patterns are converted into simple summary features before modeling.

Proposed Workplan (14 Weeks)

Phase / Weeks	Focus	Key Activities
Phase 1 — Weeks 1–2	Scoping & Literature Review	Define research question and objectives; review academic literature and existing systems/products; finalize scope with supervisor sign-off.
Phase 2 — Weeks 3–4	Usage Pattern Dataset Sourcing & Feature Engineering	Identify and source datasets (or design data-collection protocol); complete data cleaning, exploratory analysis, and ethical/privacy review where child data is involved.
Phase 3 — Weeks 5–8	Predictive Model Development (Classification on Summary Features)	Design and build the core model/system/prototype; implement baseline, then iterate; document design decisions and trade-offs.
Phase 4 — Weeks 9–11	Evaluation & Refinement	Test against defined metrics/benchmarks or original reference paper; run comparative analysis; refine based on results; conduct usability or expert validation where applicable.
Phase 5 — Weeks 12–13	Documentation	Write up methodology, results, and discussion; prepare figures/tables; complete limitations and future-work sections.
Phase 6 — Week 14	Defense Preparation	Prepare presentation and demo; rehearse defense; incorporate supervisor feedback from final review.

16. Text-Based Safety Moderation for Child-Directed Voice Assistant Transcripts

Description

Voice-enabled toys and smart speakers marketed to children raise moderation questions once their spoken interactions are converted to text. This project uses an existing, off-the-shelf speech-to-text API (e.g. Google Speech-to-Text or a similar well-documented service) to transcribe sample voice interactions, then applies standard text-classification techniques to the transcripts to flag inappropriate content — keeping the project scoped to text classification.

Prerequisites

Basic Python; comfort calling a speech-to-text API (a simple, well-documented request/response pattern); the underlying text classification uses the same scikit-learn/pretrained-classifier approach as other NLP topics in this set.

Proposed Workplan (14 Weeks)

Phase / Weeks	Focus	Key Activities
Phase 1 — Weeks 1–2	Scoping & Literature Review	Define research question and objectives; review academic literature and existing systems/products; finalize scope with supervisor sign-off.
Phase 2 — Weeks 3–4	Sample Voice Interaction Collection & Transcription Setup	Identify and source datasets (or design data-collection protocol); complete data cleaning, exploratory analysis, and ethical/privacy review where child data is involved.
Phase 3 — Weeks 5–8	Text Classification Model Development on Transcripts	Design and build the core model/system/prototype; implement baseline, then iterate; document design decisions and trade-offs.
Phase 4 — Weeks 9–11	Evaluation & Refinement	Test against defined metrics/benchmarks or original reference paper; run comparative analysis; refine based on results; conduct usability or expert validation where applicable.
Phase 5 — Weeks 12–13	Documentation	Write up methodology, results, and discussion; prepare figures/tables; complete limitations and future-work sections.
Phase 6 — Week 14	Defense Preparation	Prepare presentation and demo; rehearse defense; incorporate supervisor feedback from final review.

Business & Governance

17. Sustainable Business Models for Family Digital-Safety Platforms

Description

This project analyzes viable monetization strategies for a child-safety EdTech platform — freemium, B2B2C partnerships with schools or telecom providers, and subscription models — benchmarked against comparable EdTech business-model theses already completed in this research group. Rather than a purely technical build, the deliverable centers on a data-informed business model canvas: the student will gather comparable-company data (public financials, pricing pages, market reports) and build a simple financial projection model to stress-test each option under different adoption-rate assumptions.

Prerequisites

Business modeling/strategy; modeling skills; basic coding requirement.

Proposed Workplan (14 Weeks)

Phase / Weeks	Focus	Key Activities
Phase 1 — Weeks 1–2	Scoping & Literature Review	Review EdTech and child-safety-platform business model literature; review comparable theses in this research group; finalize research questions.
Phase 2 — Weeks 3–5	Comparable Market Research	Identify and analyze 8–12 comparable platforms (pricing, revenue model, target segment); interview or survey potential customer segments (schools, parents) if feasible.
Phase 3 — Weeks 6–9	Business Model Design & Financial Modeling	Build business model canvases for each candidate model; construct a simple financial projection spreadsheet with adoption-rate scenarios; identify key risk factors per model.
Phase 4 — Weeks 10–11	Comparative Evaluation	Score models against defined criteria (revenue potential, time-to-market, alignment with mission); present findings to supervisor for feedback.
Phase 5 — Weeks 12–13	Documentation	Write up methodology, comparative analysis, and recommendation; prepare supporting financial exhibits.
Phase 6 — Week 14	Defense Preparation	Prepare presentation; rehearse defense; incorporate supervisor feedback.

18. Regulatory Compliance Benchmarking: COPPA, GDPR-K, and Egypt's Data Protection Law for Child-Focused Platforms

Description

This project produces a comparative legal-technical analysis of what "compliant by design" architecture actually requires across three regimes relevant to localizing Play, Learn & Protect: the US COPPA framework, the EU's GDPR provisions for children's data (GDPR-K), and Egypt's Personal Data Protection Law 151/2020. Rather than a purely legal review, the deliverable translates each regime's requirements into concrete technical/architectural implications (e.g. consent-flow design, data-minimization requirements, retention limits, cross-border transfer restrictions) so the analysis is directly usable by an engineering team building the actual product.

Prerequisites

No coding requirement; strong research and legal-document analysis skills; Business Informatics or IT-governance coursework helpful.

Proposed Workplan (14 Weeks)

Phase / Weeks	Focus	Key Activities
Phase 1 — Weeks 1–2	Scoping & Regulatory Landscape Mapping	Identify all applicable regulations across target markets; define comparison framework/criteria; finalize scope with supervisor.
Phase 2 — Weeks 3–6	Regulatory Deep-Dive Analysis	Analyze each regulation's requirements for consent, data minimization, retention, parental controls, and cross-border transfer; review enforcement case studies (e.g. past COPPA fines) for practical interpretation.
Phase 3 — Weeks 7–9	Technical Translation	Translate each legal requirement into a concrete architectural/technical implication (e.g. specific consent-flow UX, data schema constraints); build a compliance-requirements matrix.
Phase 4 — Weeks 10–11	Gap Analysis & Validation	Apply the compliance matrix to a hypothetical (or actual, if available) Play Learn Protect architecture to identify gaps; validate findings against precedent industry compliance frameworks.
Phase 5 — Weeks 12–13	Documentation	Write up comparative regulatory analysis and compliance-requirements matrix; prepare an executive summary suitable for non-legal stakeholders.
Phase 6 — Week 14	Defense Preparation	Prepare presentation; rehearse defense; incorporate supervisor feedback.